

MANAN SHAH

Software Engineer | AI & Cloud Systems | U.S. Permanent Resident

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SUMMARY

Software Engineer with hands-on experience across Dialogflow CX, BigQuery, and Cloud Run on a production enterprise CCaaS platform, plus independent project work in governed Text-to-SQL pipelines and RAG-based conversational AI. Strong in REST API development, cloud observability, and production defect resolution.

EXPERIENCE

Software Engineer · Capgemini

Feb 2026 – Present

- Delivered Dialogflow CX enhancements and production defect fixes for a telecommunications client's enterprise CCaaS platform, analyzing conversational flows, tracing parameter lifecycles across 10+ pages and routes, and refining solution designs to improve routing accuracy and reduce misrouting scenarios.
- Investigated end-to-end call routing behavior across Dialogflow CX, webhook interactions, API/database calls, and cloud logs, analyzing 10,000+ call records in BigQuery to identify latency-prone steps, duplicate service calls, and flow configuration issues; contributed to optimizations that reduced average call latency by ~3 seconds per call.
- Executed backend, observability, and release validation activities using Python, Cloud Run, Cloud Logging, BigQuery, and Looker, while documenting technical decisions, testing strategy, solution trade-offs, and deployment validation through Confluence and 5+ Visio process maps.

Software Engineer · H2 Techno World

Sep 2024 – May 2025

- Designed and developed a full-stack inventory and order management platform using React.js, FastAPI, and PostgreSQL, supporting 5 core business modules: product management, inventory tracking, order processing, user access, and audit/reporting.
- Built RESTful backend services with normalized PostgreSQL schemas across 8 core entities and implemented audit/validation workflows for 4 event categories, while automating Docker-based deployments through GitHub Actions and AWS EC2 to reduce release execution from 7 manual tasks to a 3-stage deployment workflow.

PROJECTS

QueryShield AI · FastAPI · Next.js · TypeScript · SQLite · BigQuery · Gemini · SQLGlot

Jul 2026

- Designed and built a full-stack governed Text-to-SQL analytics platform enabling an end-to-end pipeline from CSV upload to AI-generated SQL, parser-based validation, controlled execution, charting, summaries, and audit logging.
- Implemented a 5+ layer query governance pipeline - JWT authentication, dataset ownership checks, SQLGlot-based SQL validation, BigQuery dry-run cost checks, and row/timeout limits to prevent unsafe or unbounded AI-generated SQL execution.
- Deployed the platform on GCP using BigQuery, Cloud Run, Cloud Storage, Secret Manager, and Artifact Registry, automating CI/CD across 15+ backend endpoints and 9 Alembic migrations via Docker, GitHub Actions, and Workload Identity Federation.

Multi-Agent Conversational AI System · Python · FastAPI · React.js · SQLite · SQLAlchemy · Cohere · FAISS

Jul 2025

- Architected a full-stack conversational AI platform using FastAPI, React.js, SQLite, Cohere, FAISS, and SQLAlchemy, supporting 4 core capabilities across RAG-based Q&A, CSV document ingestion, CRM user management, and conversation memory, with 4 frontend components for chat, user management, conversation history, and document upload.
- Built 7 RESTful API endpoints and a 6-stage RAG pipeline that retrieves prior user history, searches FAISS indexes, reranks semantic candidates, constructs contextual prompts, generates Cohere-powered responses, and logs interactions through 2 SQLite-backed CRM entities: users and conversations.

EDUCATION

M.S. Computer Science · New Jersey Institute of Technology · GPA: 3.9/4.0

Jan 2024 – May 2025

B.E. Computer Engineering · Charusat University · GPA: 9.75/10 · Top 5% of Class

Jul 2019 – May 2023

SKILLS & CERTIFICATIONS

Languages: Python, JavaScript, TypeScript, SQL, C++

Frameworks & Databases: React.js, Node.js, Next.js, FastAPI, REST APIs, SQLAlchemy, PostgreSQL, MySQL, SQLite, BigQuery

GCP: Dialogflow CX, Cloud Run, Cloud Logging, Secret Manager, Artifact Registry, Cloud Storage

AWS: EC2, S3, RDS, Lambda

AI & Data: RAG, FAISS, Cohere LLM, AWS Bedrock, Gemini, SQL Validation

Tools: Docker, GitHub Actions, GitHub, JIRA, Postman, pytest, Looker, Confluence

Certifications: AWS Cloud Practitioner · AWS Developer – Associate · Google Associate Cloud Engineer